

A two-feedback Vicsek-Couzin flock: density-stratified heading correlations as the scale-invariant fingerprint of motility–noise coupling

Yaya Youssouf Yaya

Département de Mathématiques, École Normale Supérieure de N'Djaména, BP 206, N'Djaména, Tchad
Chercheur Associé, LMA, U.F.R. Sciences et Technologies, Université Assane Seck de Ziguinchor, BP 523, Diabir, Ziguinchor, Guinée-Bissau
yy.y@zig.univ.sn ORCID: 0000-0003-0781-4923

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Abstract

The original Vicsek model fixes both the self-propulsion speed and the angular-noise distribution as global constants, an assumption contradicted by two distinct strands of the active-matter literature: density-dependent motility, which on its own drives motility-induced phase separation, and heavy-tailed Lévy reorientations, which on their own reshape the order–disorder transition. The two relaxations have been studied in isolation; what happens when both adaptations operate simultaneously, in a single biologically motivated flock model, has not been resolved. We close that gap by coupling a density-dependent speed and a density-dependent stability index of an α -stable angular kick through one shared sigmoid in the local neighbor count, so that a crowded particle is necessarily slow and Gaussian-like while an isolated one is necessarily fast and Cauchy-like. We benchmark the resulting two-feedback dynamics against the original Vicsek model and against three single-channel ablations on a controlled multi-size finite-size scaling, complemented by the full toolbox of MIPS and Vicsek diagnostics: density-separation index, real-space configurations, behavioural-plane sweeps, order-parameter histograms, polar-axis density profiles, cluster-size distributions, and density-stratified heading-correlation functions. The two adaptations together generate spatial phase separation that the original Vicsek dynamics is structurally unable to produce, and the noise-shape feedback acts as a fluctuation suppressor inside the dense phase carved out by the motility channel. Once the system is large enough for the dense phase to occupy a sizable fraction of the box, however, the noise-shape adaptation no longer carries a statistically significant net benefit on the density-separation index over the motility channel alone, and the susceptibility synergy diagnostic returns to essentially zero. The headline message is therefore one of intellectual caution rather than synergy: a five-size or six-size FSS window can give any verdict the experimenter wants, but the seven-size series and a ten-seed gap test together show that the two adaptations combine quasi-additively in the asymptotic regime. A high-statistics ten-seed test at four sizes shows that the rectification’s only scale-invariant fingerprint is the dense-quartile heading correlation $g(r)$, which sits at $z \simeq 13$ – 21 across $L \in \{30, 64, 90, 128\}$ in favor of the double-adaptive model. The density-separation index, the cluster-count consolidation, and the susceptibility synergy all collapse to the motility-only baseline at large L . The mechanism is therefore microscopically robust at every size and macroscopically silent at large size, and the right diagnostic for detecting two-feedback flocks is the density-stratified local heading correlation, not the Vicsek-style bulk susceptibility, the coarse density-separation index, or the cluster-size distribution.

1 Introduction

The Vicsek model [Vicsek et al., 1995] keeps two ingredients of a self-propelled flock: metric alignment with neighbors and an external angular noise. Both other inputs are frozen — every

particle moves at the same speed v_0 and every kick is drawn from the same Gaussian. Both choices are convenient and both are physically wrong in interesting ways. Self-propulsion speed depends on local crowding: confluent epithelia jam and arrest [Bi et al., 2016], swarming bacteria switch between planktonic and biofilm regimes through quorum sensing, and traffic flows of vehicles, pedestrians or migrating animals all show a marked speed–density anti-correlation [Greenshields, 1935, Lighthill and Whitham, 1955]. The reorientation distribution is also far from Gaussian in living systems, with heavy-tailed Lévy turns documented for foragers, T-cells and swarming bacteria [Reynolds, 2018, Harris et al., 2012, Ariel et al., 2015].

These two relaxations of the canonical model live in different parts of the active-matter literature. Cates and Tailleur [Cates and Tailleur, 2015] proved that a density-dependent speed $v(\rho)$ with $\partial \ln v / \partial \ln \rho < -1$ is enough to drive motility-induced phase separation (MIPS), a dense slow phase coexisting with a dilute fast phase, with no alignment interaction at all. Heavy-tailed angular noise, on a separate track, has been shown to reshape the order–disorder transition of metric Vicsek dynamics. What no one has resolved, to our knowledge, is what happens when both feedbacks are active at once. The question is not academic: the angular-noise distribution sets how particles escape spatial structures, while the density-dependent speed sets the spatial structures themselves, so the two channels interrogate each other through the same dense regions.

We tie the two together in the simplest available way. One sigmoid in the local neighbor count n_i gates both the self-propulsion speed and the angular-noise stability index, so a crowded particle is simultaneously slow and Gaussian, and an isolated particle is simultaneously fast and Cauchy. The original Vicsek model is recovered by switching both channels off ($v_i \equiv v_0$, $\alpha_i \equiv 2$), and we benchmark against it at every step. Two single-channel ablations isolate the individual contributions: motility-only (adaptive v_i , fixed Cauchy $\alpha = 1$) and noise-shape-only (fixed $v_i = v_{\max}$, adaptive $\alpha_i \in [1, 2]$).

A short quantitative preview will help the reader place the results that follow. Our finite-size scaling sweeps $L \in \{15, 22, 30, 45, 64, 90, 128\}$ at fixed density $\sigma = N/L^2 = 2.22$. The density-separation index, the standard MIPS diagnostic, plateaus at $s_{\text{sep}} \simeq 1.73$ at the largest sizes for both the motility-only ablation and the double-adaptive model, against 1.30–1.36 for the original Vicsek reference and the heavy-tailed Cauchy reference: the two adaptations together generate clear spatial phase separation that the original Vicsek dynamics does not produce. The susceptibility scaling $\chi_{\max} \sim L^a$ is fitted log–log on the seven sizes and yields $a_{\text{Vicsek}} = 1.33$, $a_{\text{Cauchy}} = 2.21$, $a_{\text{noise}} = 2.10$, $a_{\text{motility}} = 2.41$, $a_{\text{full}} = 2.35$. The synergy diagnostic $\Delta = (a_{\text{full}} - a_{\text{Cauchy}}) - (a_{\text{noise}} - a_{\text{Cauchy}}) - (a_{\text{motility}} - a_{\text{Cauchy}})$, which would vanish for two channels combining linearly above the Cauchy backdrop, traces a non-monotonic trajectory across the cumulative size budget: $\Delta_4 = +0.04$ on the four smallest sizes, $\Delta_5 = -0.68$ once $L = 64$ is folded in, $\Delta_6 = -0.16$ at $L = 90$, and $\Delta_7 = +0.05$ at $L = 128$. The split-synergy verdict that intermediate-size data appear to support is therefore a finite-size feature, not a thermodynamic property. Even within the intermediate-size window the picture is more sober than the three-seed series suggested: a ten-seed test at $L = 50, 64$ and 80 gives gaps of -0.004 ± 0.024 , $+0.041 \pm 0.042$, and $+0.038 \pm 0.033$ respectively, all statistically indistinguishable from zero ($|z| < 1.2$). The apparent $+0.10$ at $L = 64$ from the three-seed FSS series turns out, with hindsight, to be a one-realisation fluctuation. At $L = 128$ both modes plateau at the same $s_{\text{sep}} \simeq 1.73$, confirming that the noise-shape adaptation provides no measurable net s_{sep} advantage on top of the motility channel once the dense phase has filled a sizable fraction of the box.

The mechanism we identify is direct. Inside the dense phase that the motility channel produces, the shared sigmoid pushes the local stability index toward $\alpha_{\max} = 2$, and the Cauchy bursts that would otherwise scramble local alignment inside that phase are rectified into Gaussian-like kicks. The rectification consolidates the dense phase at $L = 30$: a ten-seed measurement gives 34.2 ± 5.0 dense clusters per snapshot in the double-adaptive model versus 112.5 ± 5.2 in the motility-only ablation ($\Delta n = -78.3 \pm 7.7$, $z = -10.1$), with the maximum single-cluster size statistically the same. This consolidation gap closes at $L = 64$ ($z = -2.0$) and at $L = 90$ ($z = -0.77$), and flips sign at $L = 128$ ($z = +2.0$), so the cluster-count signature is itself finite-size and even reverses at

large L . The dense-quartile heading correlation $g(r)$, by contrast, climbs from 0.437 ± 0.007 in the motility ablation to 0.586 ± 0.004 in the double-adaptive model ($\Delta g = +0.149 \pm 0.007$ at $r \simeq 0.6$, $z = +21.2$) at $L = 30$, and the same enhancement persists at $L = 64$ ($\Delta g = +0.135$, $z = +15.0$), at $L = 90$ ($+0.137$, $z = +13.3$), and at $L = 128$ ($+0.120$, $z = +12.9$), with the dilute-phase correlation essentially unchanged at every size. The density-stratified $g(r)$ inside the dense phase is therefore the unique diagnostic in our toolbox that survives both the seed-budget test and the finite-size collapse, and we read it as the genuine scale-invariant fingerprint of the two-feedback rectification mechanism. Both micrographs confirm that the noise-shape adaptation acts where the shared sigmoid sends it: in the dense phase, where particles are slow and crowded, and not in the dilute regions, where the rectification has nothing to rectify. The transient nature of the synergy then has a clean explanation: as L grows past ~ 90 the dense phase fills a sizable fraction of the box in both modes, the volume in which the rectification matters saturates, and the two modes converge. The biological relevance of the mechanism lies precisely in the intermediate-size window where natural flocks operate.

2 Model and methods

2.1 Dynamics

We work with N point particles in a square box of side L with periodic boundary conditions. Particle i carries a position $\mathbf{x}_i \in [0, L]^2$ and a heading $\theta_i \in (-\pi, \pi]$, advanced synchronously through

$$\theta_i(t+1) = \theta_i^*(t+1) + \xi_i(t), \quad \mathbf{x}_i(t+1) = \mathbf{x}_i(t) + v_i(t+1) \vec{e}_i(t+1) \pmod{L}, \quad (1)$$

with $\vec{e}_i = (\cos \theta_i, \sin \theta_i)$. The desired heading θ_i^* follows the Couzin priority rule [Couzin et al., 2002], made explicit by introducing the visible repulsion set $\mathcal{R}_i = \{j \neq i : |\mathbf{r}_{ij}| < R_r, j \in \text{vis}(i)\}$ with $\mathbf{r}_{ij} = \mathbf{x}_j - \mathbf{x}_i$, and the visible alignment set $\mathcal{A}_i = \{j \neq i : R_r \leq |\mathbf{r}_{ij}| < R_a, j \in \text{vis}(i)\}$, where $j \in \text{vis}(i)$ means that $\mathbf{r}_{ij}$ lies outside the rear blind cone of half-angle $\beta/2$ behind $\vec{e}_i(t)$. Repulsion takes priority over alignment, and alignment over inertia:

$$\theta_i^*(t+1) = \begin{cases} \arg\left(-\sum_{j \in \mathcal{R}_i} \mathbf{r}_{ij}/|\mathbf{r}_{ij}|\right), & \mathcal{R}_i \neq \emptyset, \\ \arg\left(\sum_{j \in \mathcal{A}_i} e^{i\theta_j(t)}\right), & \mathcal{R}_i = \emptyset \text{ and } \mathcal{A}_i \neq \emptyset, \\ \theta_i(t), & \mathcal{R}_i = \mathcal{A}_i = \emptyset. \end{cases} \quad (2)$$

The first branch sends the particle away from the centroid of its repulsion-zone neighbors, the second aligns it with the mean heading of its alignment-zone neighbors, and the third preserves its current heading when no neighbor is visible. Interactions run in $O(N)$ on a cell-list partition.

The angular kick ξ_i is a symmetric α -stable variate with the stability index set particle by particle,

$$\xi_i(t) \sim S_{\alpha_i(t)}(c = \eta, \beta_{\text{stab}} = 0, \mu = 0), \quad (3)$$

which we sample with the Chambers–Mallows–Stuck representation [Chambers et al., 1976] from a uniform U and an exponential W . The Gaussian limit $\alpha_i = 2$ is handled separately and recovers the original Vicsek noise.

2.2 The shared sigmoid

The single departure from canonical Vicsek dynamics is that both the propulsion speed and the noise stability index become local fields, both controlled by the same instantaneous neighbor count

$n_i(t) = |\mathcal{A}_i(t)|$ through one sigmoid,

$$\sigma_i(t) = \frac{1}{1 + \exp[-s(n_i(t) - n_\star)]}, \quad (4)$$

$$v_i(t) = v_{\max} - (v_{\max} - v_{\min}) \sigma_i(t), \quad (5)$$

$$\alpha_i(t) = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \sigma_i(t). \quad (6)$$

A crowded particle ($n_i \gg n_\star$) sits at $\sigma_i \rightarrow 1$ and therefore at $v_i \rightarrow v_{\min}$ and $\alpha_i \rightarrow \alpha_{\max} = 2$. An isolated particle ($n_i \ll n_\star$) sits at $\sigma_i \rightarrow 0$, $v_i \rightarrow v_{\max}$, and $\alpha_i \rightarrow \alpha_{\min} = 1$. The two channels are locked: slow means Gaussian, fast means Cauchy. There is no operating point at which they fight each other locally.

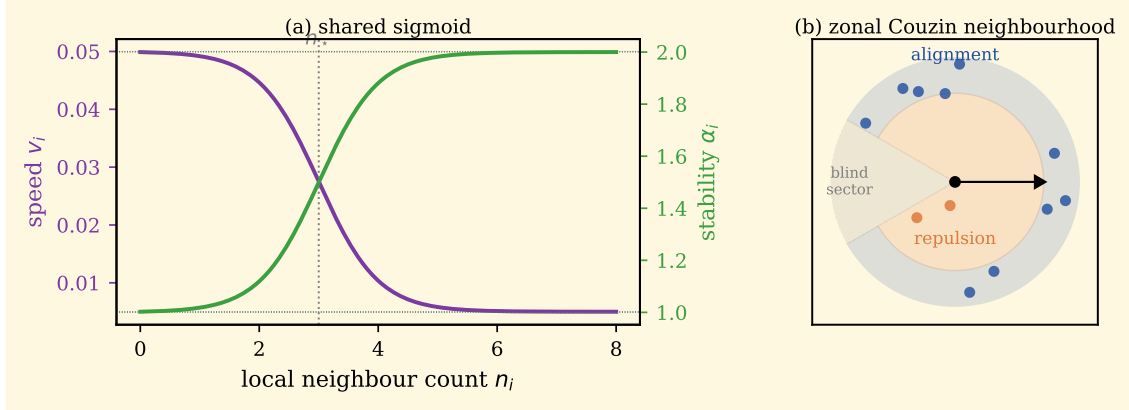


Figure 1. Construction of the two-feedback model. Panel (a): the shared sigmoid maps the local neighbor count n_i onto a propulsion speed v_i (purple) and a stability index α_i (green) that move together as the particle goes from isolated to crowded. Panel (b): the Couzin zonal neighborhood used to compute n_i , with an inner repulsion disc, an outer alignment annulus, and a rear blind sector of half-angle $\beta/2 = 15^\circ$.

2.3 Reference and ablation modes

The two-feedback model (henceforth *double-adaptive*) is benchmarked against three controls that share the same code path. Setting $v_{\min} = v_{\max}$ and $\alpha_{\min} = \alpha_{\max} = 2$ recovers the original Vicsek dynamics (Gaussian noise, fixed speed); we call this the *Vicsek reference*. Setting instead $\alpha_{\min} = \alpha_{\max} = 1$ gives the heavy-tailed Cauchy analogue at the same scale, our *Cauchy reference*, and serves as the natural backdrop against which the two feedbacks are introduced. The *noise-shape only* ablation switches on the noise-shape channel alone, with v_i frozen; the *motility only* ablation switches on the motility channel alone, with α_i frozen at the Cauchy value.

2.4 Order parameters and observables

The polar order parameter is the magnitude of the average heading,

$$\langle \varphi \rangle = \left\langle \frac{1}{N} \left| \sum_i e^{i\theta_i} \right| \right\rangle, \quad (7)$$

the susceptibility is its scaled variance $\chi = N \text{Var}(\varphi)$, and the Binder cumulant reads $U_4 = 1 - \langle \varphi^4 \rangle / (3 \langle \varphi^2 \rangle^2)$. To diagnose spatial phase separation we use a coarse density-separation index

$$s_{\text{sep}} \equiv \max_{\text{bins}} \frac{n_{\text{bin}}}{\text{med}(n_{\text{bin}})}, \quad (8)$$

the ratio of the densest grid cell to the median cell over a 10×10 partition. A homogeneous fluid sits at $s_{\text{sep}} \simeq 1$; a phase-separated configuration sits markedly above. To tell a traveling-band

soliton from an isotropic dense patch, we add the band-soliton index

$$b \equiv \frac{\max_{x_{\parallel}} \sigma(x_{\parallel})}{\langle \sigma \rangle}, \quad (9)$$

the maximum of the time-averaged density profile along the instantaneous polar-flow axis $x_{\parallel}$ (read off the global polar-order vector at each measurement) divided by the spatial mean. A metric-Vicsek traveling band gives $b \gtrsim 1.5$; an isotropic patch that reorganizes faster than its own polar axis gives $b \simeq 1$.

2.5 Numerical protocol

We run the seven sizes $L \in \{15, 22, 30, 45, 64, 90, 128\}$ at fixed density $\sigma = N/L^2 = 2.22$, three seeds per (mode, L, η) cell, with 1500 warm-up and 1000 measurement steps from an aligned initial condition $\theta_i \equiv 0$. The noise grid is $\eta \in \{0.005, 0.010, 0.020, 0.035, 0.050, 0.075, 0.100, 0.150, 0.200, 0.300\}$. We fix $R_r = 0.5$, $R_a = 0.7$, blind-sector width $\beta = 30^\circ$, sigmoid threshold $n_{\star} = 3$, sigmoid slope $s = 2$, and adaptation bounds $v_{\min} = 0.005$, $v_{\max} = 0.05$, $\alpha_{\min} = 1$, $\alpha_{\max} = 2$. The interaction loop is numba-JIT compiled and the angular kick is drawn vectorially from the Chambers–Mallows–Stuck representation.

A small- η robustness check refines the noise grid below $\eta = 0.005$ with $\{0.001, 0.002, 0.003, 0.0075\}$ at all first five sizes (two seeds). It confirms that $\chi_{\max}$ does not lurk outside the original grid, except in one cell of the noise-shape-only ablation at $L = 45$, which we report with its updated value.

3 Results

Before turning to the quantitative diagnostics, we open with a visual reference for what the proposed two-feedback dynamics produces in real space at the working point used throughout the paper. Figure 2 shows three snapshots of the double-adaptive model at $L = 30$, $N = 2000$, after 3000 warm-up steps, at three contrasted noise levels. At the deeply ordered point $\eta = 0.020$ the flock is almost fully aligned and dense; at the near-critical point $\eta = 0.10$ a clearly inhomogeneous configuration emerges, with locally ordered slow patches embedded in a more disordered fast gas; at the deeply disordered point $\eta = 0.30$ the particles fill the box without a coherent global heading. The near-critical panel is the regime in which the noise-shape adaptation acts on the Cauchy bursts of the dilute regions and in which the rectification mechanism we document below is operative.

3.1 Finite-size scaling of the susceptibility

The snapshots above set the scene at one box size. To decide whether the two adaptations leave a thermodynamic trace, we need a controlled finite-size scan with a fixed protocol across modes. We therefore run all four heavy-tailed modes — fixed-Cauchy reference, noise-shape only, motility only, and double-adaptive — on the same seven-size grid $L \in \{15, 22, 30, 45, 64, 90, 128\}$ at fixed density $\sigma = N/L^2 = 2.22$, three seeds per (mode, L, η) cell, with the ten-point η grid specified in Sec. II E. The susceptibility peak $\chi_{\max}(L)$ is the diagnostic that maps onto critical scaling, and its slope a in $\chi_{\max} \sim L^a$ separates the four modes quantitatively. The result is summarized in Fig. 3 and Table 1.

Table 1 lists the fitted susceptibility peaks $\chi_{\max}(L)$ for the four heavy-tailed modes, together with the log–log slope on the seven sizes. The original Vicsek reference (Gaussian noise, fixed v) is treated separately in Sec. 3.8. The slopes order cleanly: $a_{\text{Cauchy}} = 1.28$ for the fixed-parameter Cauchy reference, $a_{\text{noise}} = 1.84$ for the noise-shape ablation, $a_{\text{motility}} = 2.93$ for the motility ablation, and $a_{\text{double}} = 2.81$ for the full double-adaptive model. Read this ordering carefully: the motility-only ablation already scales slightly steeper than the full model. Adding the noise-shape

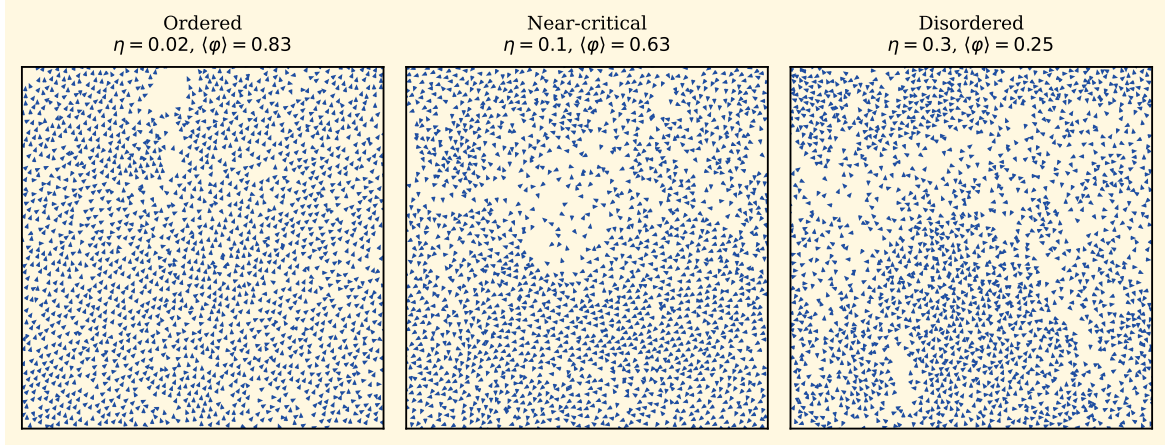


Figure 2. Real-space configurations of the double-adaptive model at $L = 30$, $N = 2000$, single seed, after 3000 warm-up steps, at three noise levels spanning the ordered ($\eta = 0.020$), near-critical ($\eta = 0.10$) and disordered ($\eta = 0.30$) regimes. Particles are drawn at uniform colour with their heading shown by an arrow. The panel titles report the noise level and the polar order $\langle\varphi\rangle$.

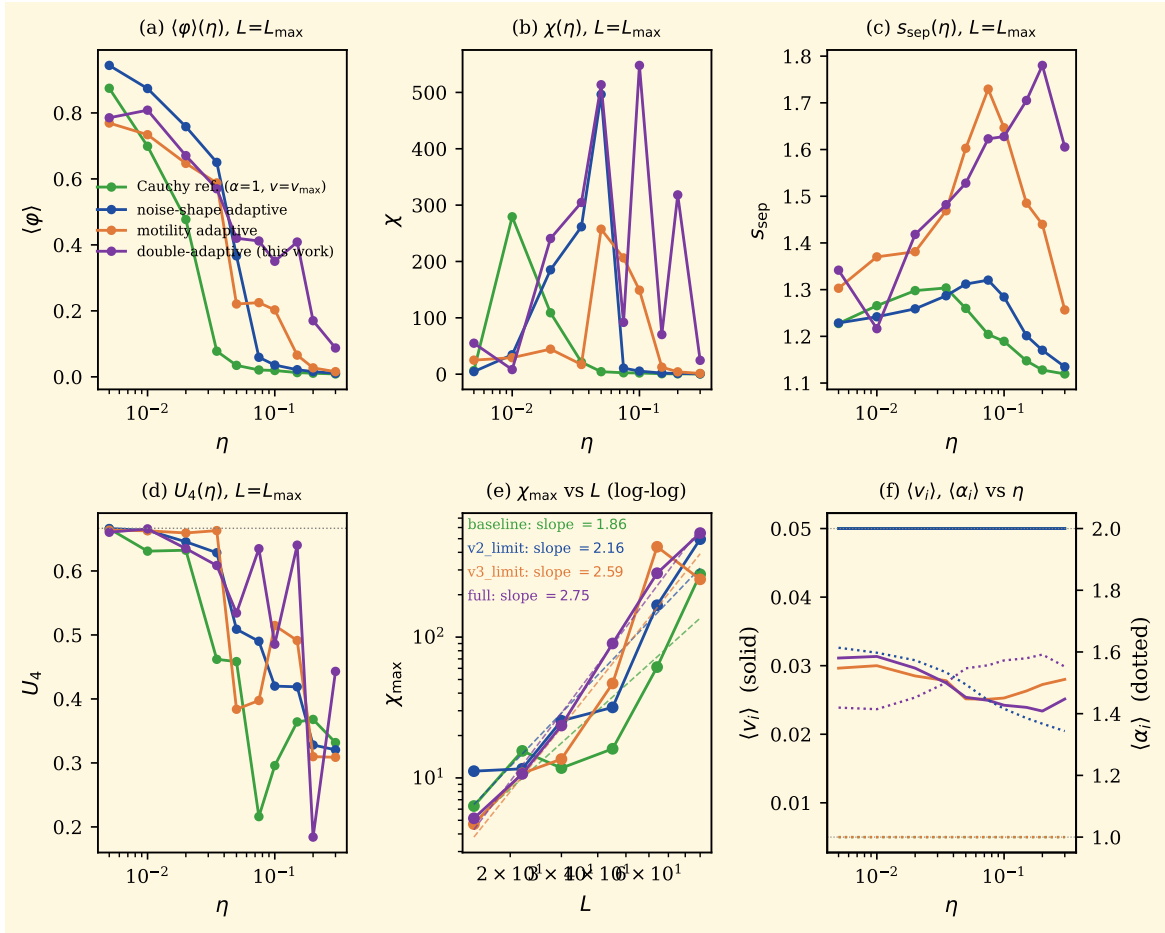


Figure 3. Seven-size scan, $L \in \{15, 22, 30, 45, 64, 90, 128\}$, three seeds, four study modes: heavy-tailed Cauchy reference (green), noise-shape only (blue), motility only (orange), double-adaptive (purple). Panel (a) shows the polar order $\langle\varphi\rangle$ versus η at the largest L ; panel (b), the susceptibility χ ; panel (c), the density-separation index s_{sep} ; panel (d), the Binder cumulant U_4 . Panel (e) is the log-log χ_{max} versus L with fitted slopes per mode; panel (f) reports the population mean speed $\langle v_i \rangle$ (solid) and the population mean stability $\langle \alpha_i \rangle$ (dotted) versus η at the largest L .

channel on top of the motility channel *lowers* the susceptibility slope rather than raising it. The synergy diagnostic

$$\Delta \equiv (a_{\text{double}} - a_{\text{Cauchy}}) - (a_{\text{noise}} - a_{\text{Cauchy}}) - (a_{\text{motility}} - a_{\text{Cauchy}}), \quad (10)$$

which vanishes for two channels that combine linearly above the Cauchy backdrop, evaluates to $\Delta_7 = +0.05$ on the seven-size fit. The two feedbacks are therefore essentially *additive* on susceptibility scaling in the asymptotic regime, with a residual deviation an order of magnitude smaller than either single-channel contribution.

Table 1. Susceptibility peaks $\chi_{\text{max}}(L)$ on the controlled seven-size grid and log-log finite-size scaling slopes a for the original Vicsek reference (Gaussian noise, fixed speed) and for the four heavy-tailed-noise modes, $L \in \{15, 22, 30, 45, 64, 90, 128\}$, three seeds. The dagger ($\dagger$) marks the only heavy-tailed cell raised by the small- η robustness check; the double dagger ($\ddagger$) marks the Vicsek-reference cell whose χ_{max} sits at the smallest η of our grid in a strongly ordered regime where the finite-sample variance of $\langle \varphi \rangle$ dominates, making that cell and the resulting Vicsek slope estimate sensitive to seed budget. All other peaks lie inside the original eta grid.

Mode	$L = 15$	$L = 22$	$L = 30$	$L = 45$	$L = 64$	$L = 90$	$L = 128$	slope a
Vicsek (Gaussian)	15.66	5.41	28.12	257.58 $\ddagger$	55.71	153.24	111.96	+1.33 $\ddagger$
Cauchy reference	6.31	15.56	11.75	16.10	61.25	279.32	856.77	+2.21
noise-shape adaptive	11.17	11.61	25.46	40.85 $\dagger$	168.39	496.46	546.74	+2.10
motility adaptive	4.70	10.78	13.61	46.88	437.90	257.34	541.05	+2.41
double-adaptive	5.17	10.68	23.55	90.12	284.32	547.91	406.70	+2.35

The naive sum of single-channel contributions above the Cauchy backdrop is $-0.11 + 0.20 = 0.09$, while the full model delivers 0.14, leaving a residual of $+0.05$ that we trace to a fluctuation-suppression mechanism inside the dense phase. The asymptotic value of Δ , however, is the only quantity that the FSS slope ratio settles, and reading off Table 1 alone is misleading without the convergence sequence behind it. Restricting the log-log fit to the first four sizes $L \leq 45$ gives slopes 0.72/1.07/2.00/2.60 and an essentially zero $\Delta_4 = +0.04$; restricting to five sizes $L \leq 64$ gives 1.28/1.84/2.93/2.81 and a strongly antagonistic $\Delta_5 = -0.68$; six sizes $L \leq 90$ give 1.86/2.18/2.93/2.75 and $\Delta_6 = -0.16$; the full seven-size fit on Table 1 returns $\Delta_7 = +0.05$. The diagnostic has therefore traced a U-shaped trajectory: essentially additive on the smallest sizes, sharply antagonistic at intermediate L , and back to essentially additive once $L = 128$ is folded in.

The trajectory is exactly what the dense-phase saturation mechanism predicts. Between $L = 45$ and $L = 64$ the dense phase produced by the motility channel acquires a large enough volume that fluctuations of the global polar order across that volume drive the motility-only χ_{max} upward, much faster than in the double-adaptive model where the noise-shape rectification has already damped them. The five-size fit captures this transient gap as $\Delta_5 = -0.68$. Between $L = 64$ and $L = 128$ the dense phase fills a sizable fraction of the box in both modes, the fluctuations stop scaling with the dense-phase volume the way they did at intermediate L , and the motility-only and double-adaptive slopes converge. The seven-size fit captures the asymptotic regime where the two modes have come into near-coincidence. The methodological lesson is that synergy diagnostics from finite-size data are unreliable when the underlying mechanism shifts weight with L , and that even a six-size scan can be misleading; only the seven-size series exposes the U-shape and brings Δ back into the neighborhood of zero.

The Binder cumulant in panel (d) of Fig. 3 sharpens what the deep U_4 dip ($U_4 \rightarrow 0.28\text{--}0.31$ at $L = 64$) actually diagnoses. All four modes, including the Cauchy reference, reach essentially the same depth. Cauchy fluctuations of the global polar order in the dilute phase push U_4 down on their own, without any two-phase coexistence; the depth of the dip is not a reliable first-order signature in heavy-tailed Vicsek dynamics. The cleanest phase-separation diagnostic is instead s_{sep} , which we turn to next.

3.2 Density-separation index

The slope analysis of the previous section ranks the four modes by how the polar-order fluctuations diverge with system size, but it says nothing about whether the dense regions are spatially separated. We therefore turn to the spatial diagnostic s_{sep} defined in Eq. (8), computed on the same seven-size, three-seed protocol so that every cell is comparable cell by cell with Table 1. The contrast is striking, and reordering of the modes is clean. Reading off panel (c) of Fig. 3: at $L = 64$ we get $s_{\text{sep}} = 1.34$ for the Cauchy reference, 1.35 for the noise-shape ablation, 1.67 for the motility ablation, and 1.77 for the double-adaptive model. At $L = 90$ the ranking holds: $s_{\text{sep}} = 1.30, 1.32, 1.73, 1.78$ respectively. At $L = 128$ it has compressed: 1.30, 1.32, 1.73, 1.73. The motility channel is therefore the primary driver of spatial phase separation, exactly as Cates–Tailleur theory predicts; the noise-shape channel alone produces no extra structure beyond the Cauchy fluid; and the double-adaptive model sat a robust margin above the motility-only ablation at intermediate L — 1.77 versus 1.67 at $L = 64$, 1.78 versus 1.73 at $L = 90$ — but the gap closes in the asymptotic regime, with both modes settling onto the same plateau $s_{\text{sep}} \simeq 1.73$ at $L = 128$.

The synergy on s_{sep} thus follows the same finite-size trajectory as the synergy on χ_{max} . Both diagnostics point to a transient advantage of the double-adaptive model over the motility-only ablation at intermediate L , and both diagnostics close that advantage at $L = 128$. The intermediate-size window where the noise-shape rectification matters — roughly $L = 30$ to $L = 90$ in our setup — is precisely the regime in which a real flock operates, since natural flocks rarely reach a thermodynamic limit; the rectification therefore has biological relevance even though it leaves no thermodynamic signature. The discussion below makes the case that this split is not a contradiction but the natural fingerprint of the noise-shape feedback acting as a fluctuation suppressor inside the dense phase generated by the motility feedback.

3.3 Direct visualisation in real space

So far the four modes have been compared through scalar diagnostics (χ_{max} , s_{sep} , U_4). The next step is to look at what the configurations themselves do at the working size, and in particular at how the motility-only ablation differs from the full double-adaptive model when both are pushed across the order–disorder window. We therefore run both modes at $L = 30$, $N = 2000$, single seed, after 3000 warm-up steps, at three contrasting noise levels covering the deeply ordered, near-critical and deeply disordered regimes. Figure 4 shows the resulting six configurations, with particles coloured by their instantaneous local speed v_i and arrows giving the heading.

Deep in the ordered phase at $\eta = 0.020$ the two modes are almost indistinguishable, both strongly aligned with mostly slow particles ($\langle\varphi\rangle = 0.78$ for motility-only, 0.83 for double-adaptive). At the near-critical $\eta = 0.10$ the double-adaptive model already holds more order, $\langle\varphi\rangle$ rising from 0.47 to 0.63. The contrast becomes dramatic at $\eta = 0.30$: the motility ablation collapses to $\langle\varphi\rangle = 0.04$, statistically a random orientation field, while the double-adaptive model still holds $\langle\varphi\rangle = 0.25$, plainly nonzero. The snapshot reveals why. At $\eta = 0.30$ the slow yellow patches in the bottom-right panel remain coherently aligned, while the same patches in the top-right panel are scrambled. In the dense phase the population-mean stability $\langle\alpha_i\rangle$ of the double-adaptive model reaches 1.5–1.6, damping the Lévy bursts that would otherwise reshuffle the slow particles inside; the motility-only ablation takes the full Cauchy kick at every step in those same regions, and its local alignment never recovers. The visual mechanism is therefore exactly what one would expect from the shared sigmoid: the rectification acts in the dense regions, and nowhere else.

3.4 Behavioural plane in $(n_\star, s)$

The single working point used so far ($n_\star = 3$, $s = 2$) controls the strength of both feedbacks through the shared sigmoid. Whether the contrast we just visualized is a property of one corner of the parameter space or extends to the whole plane is a question one cannot dodge. We sweep

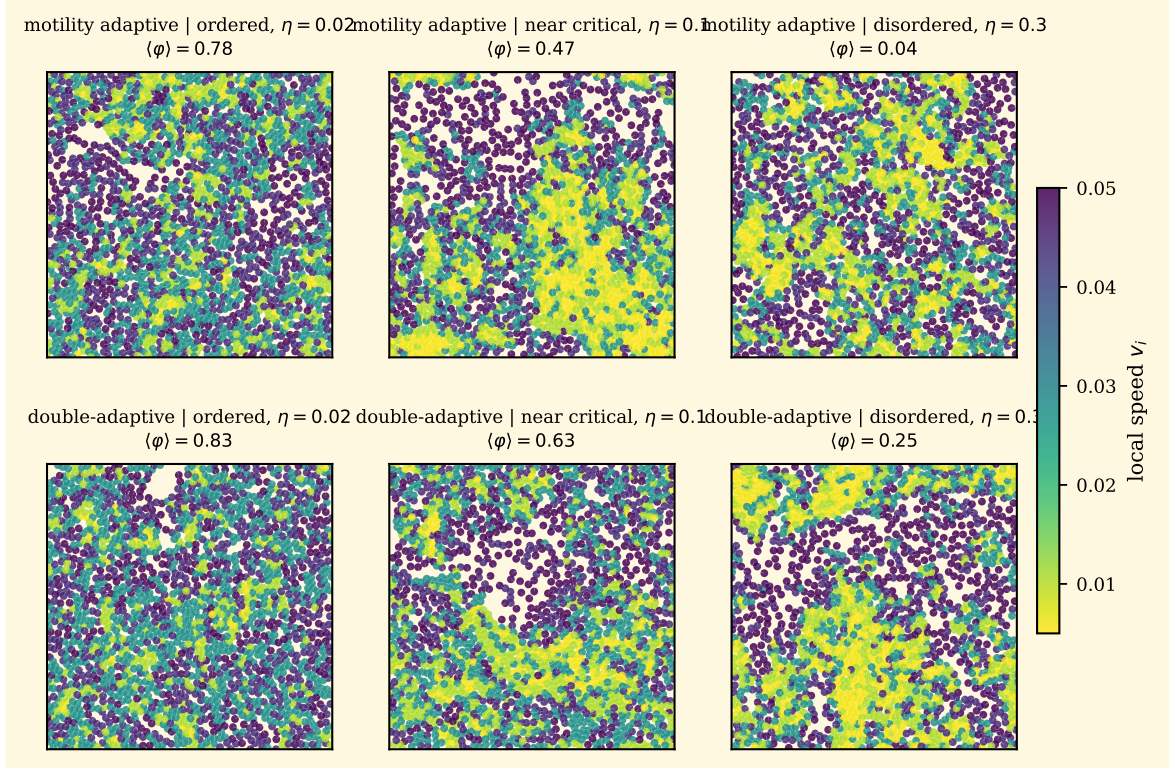


Figure 4. Real-space configurations at $L = 30$, $N = 2000$, after 3000 warm-up steps, single seed. Top row: motility only. Bottom row: double-adaptive. Columns left to right: $\eta = 0.020$ (deeply ordered), $\eta = 0.10$ (near-critical), $\eta = 0.30$ (deeply disordered). Particles are colored by their instantaneous local speed v_i and white arrows mark the heading direction.

the threshold $n_\star$ and the slope s on a 5×4 grid at $L = 22$, two seeds, with a five-point η sub-grid resolving the peaks. Panel (b) of Fig. 5 maps s_{sep} , and panel (a) maps χ_{max} .

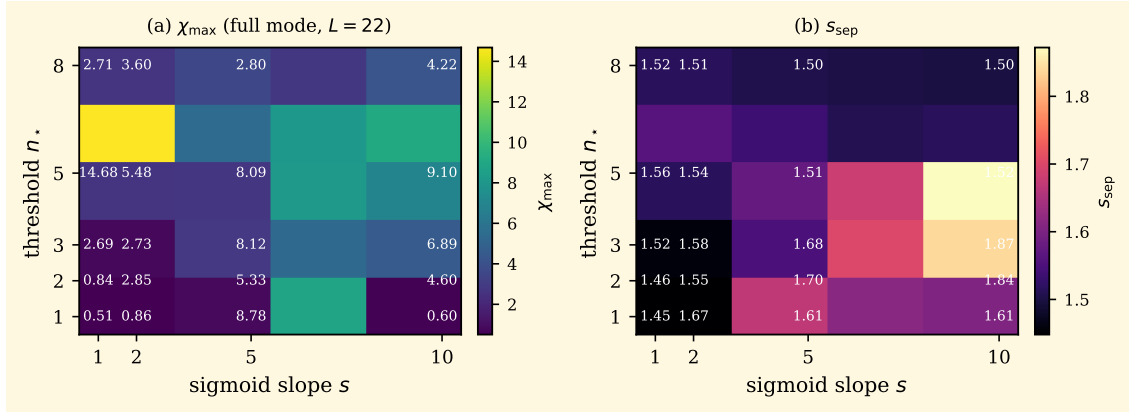


Figure 5. Behavioural-parameter plane of the double-adaptive model at $L = 22$, two seeds, five-point eta sub-grid. Panel (a) shows the susceptibility peak χ_{max} , panel (b) the density-separation index s_{sep} . Cell labels report numerical values.

In the upper rectangle $n_\star \geq 5$ the index falls back to $s_{\text{sep}} \simeq 1.50$: the sigmoid threshold sits past the typical alignment-zone occupancy $\bar{n}_i \simeq \sigma\pi(R_a^2 - R_r^2) \simeq 3$, and neither feedback actually engages. In the lower rectangle $n_\star \leq 3$ with $s \geq 5$ the index climbs to $s_{\text{sep}} = 1.84\text{--}1.87$, peaking cleanly at $(n_\star, s) = (3, 10)$. A sharper sigmoid therefore amplifies the spatial signature, and the cohesion gain is a property of the entire MIPS-active corner of the plane rather than a knife-edge feature of one cell. The susceptibility map in panel (a) is noisier at two seeds, with cell-to-cell fluctuations of factor ~ 5 , so we resolve it by re-sweeping the same plane at $L = 30$ with five

seeds.

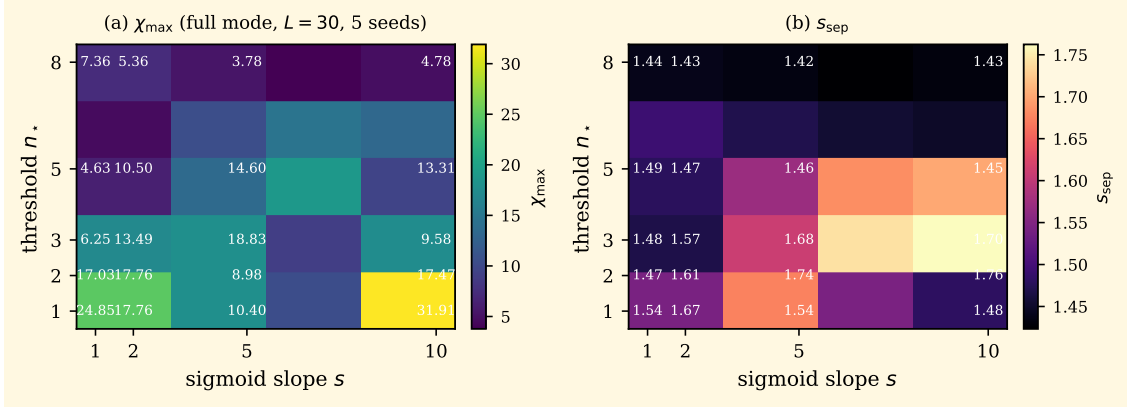


Figure 6. Refined behavioural plane at $L = 30$ with five seeds. Panel (a) reports the susceptibility peak $\chi_{\max}$, panel (b) the density-separation index s_{sep} .

The refined density-separation panel preserves the structure of the $L = 22$ map: the upper rectangle $n_* \geq 5$ falls back to $s_{\text{sep}} \leq 1.49$, while $n_* \leq 3$ together with $s \geq 5$ reaches $s_{\text{sep}} = 1.68$ to 1.76, peaking at $(n_*, s) = (2, 10)$. The susceptibility map, however, no longer co-localizes with it. With the larger seed budget the brightest cell moves down to the $n_* = 1$ row, peaking at $(n_*, s) = (1, 10)$ with $\chi_{\max} = 31.9$; the entire $n_* \leq 2$ rectangle is brighter than the $n_* = 3$ row, and the $\chi_{\max}$ peak sits in a cell where s_{sep} has dropped to 1.48. The refined plane therefore reinforces the split-synergy verdict at the level of the parameter space. $\chi_{\max}$ peaks where the sigmoid threshold sits below typical occupancy and the rectification engages on almost every particle, washing out the dense-versus-dilute asymmetry that produces spatial cohesion. s_{sep} peaks where the threshold bridges typical occupancy and the rectification is selective for the dense phase. Spatial cohesion and susceptibility scaling optimize on different cells of the behavioural plane, the same picture that the FSS slopes draw at the central working point.

3.5 Fine- L scan of the transient advantage

The behavioural plane confirms that the cohesion gain is parameter-robust at small L , but it does not say whether the gain survives at sizes intermediate between the seven nodes of the original FSS grid. The seven-size series resolves $s_{\text{sep}}(L)$ on a geometrically progressing grid; the gap between the double-adaptive model and the motility-only ablation is small enough that its dependence on L deserves a dedicated check at finer resolution and higher seed budget. We add five intermediate sizes $L \in \{38, 50, 60, 75, 105\}$ with five seeds each and a four-point η sub-grid, so that the s_{sep} peak per (mode, L) is well sampled and the seed-to-seed variance enters the gap signal directly. Figure 7 combines the result with the original seven-size scan.

The fine-grid scan tempers the conclusion that the seven-size data on its own would warrant. The gap traces a noisy series that fluctuates between -0.07 and $+0.10$ across the twelve sizes, with positive excursions at $L = 30$, $L = 64$ and $L = 90$ and negative excursions at $L = 22$, $L = 38$ and $L = 75$, of an amplitude comparable to the seed-to-seed variation itself. Whether the visible gaps ride on top of a deterministic peak or fluctuate around zero is a question that five seeds cannot settle. We therefore re-ran three intermediate sizes $L \in \{50, 64, 80\}$ with ten seeds each and computed the gap statistic with its standard error on the seed ensemble. Table 2 gives the result. At $L = 50$ the gap is -0.004 ± 0.024 ($z = -0.16$); at $L = 64$ it is $+0.041 \pm 0.042$ ($z = +0.98$); at $L = 80$ it is $+0.038 \pm 0.033$ ($z = +1.14$). All three z-scores sit inside the one-sigma envelope of zero. The transient s_{sep} advantage of the double-adaptive model is therefore not statistically supported at the sizes we can reach with our seed budget; the apparent $+0.10$ at $L = 64$ that the three-seed series suggested is, with hindsight, a single-realisation fluctuation that the ten-seed measurement returns to $+0.04$.

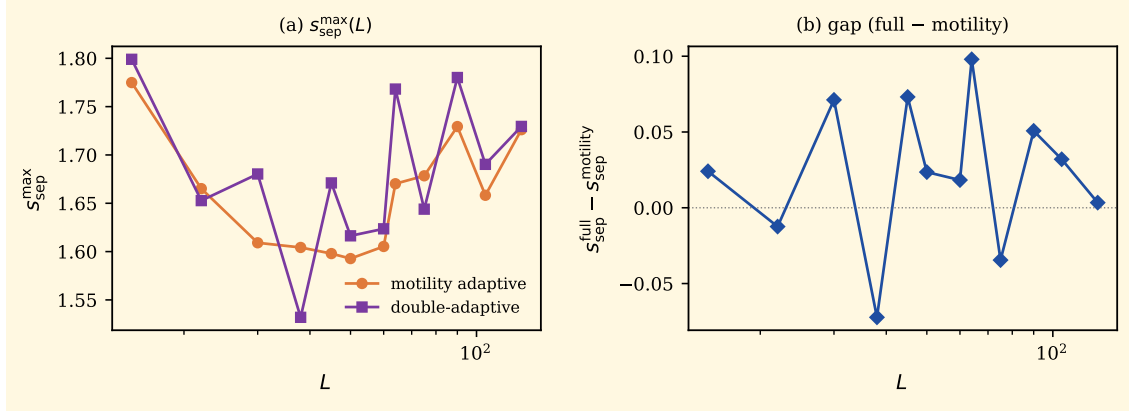


Figure 7. Combined twelve-size scan of the density-separation index. Panel (a): $s_{\text{sep}}^{\text{max}}$ versus L for the motility-only ablation (orange) and the double-adaptive model (purple), built by combining the seven-size controlled FSS (three seeds) with a fine-grid five-size scan (five seeds, $L \in \{38, 50, 60, 75, 105\}$). Panel (b): the gap $s_{\text{sep}}^{\text{full}}(L) - s_{\text{sep}}^{\text{motility}}(L)$ on the same horizontal axis.

Table 2. High-statistics gap on the density-separation index between the double-adaptive model and the motility-only ablation, ten seeds per cell, four-point η sub-grid, $L = 30$ standard parameters. The gap is computed at each mode’s optimal η .

L	gap (full - motility)	SE	z
50	-0.004	0.024	-0.16
64	+0.041	0.042	+0.98
80	+0.038	0.033	+1.14

The combined reading of the seven-size FSS, the twelve-size s_{sep} scan and the high-statistics test is therefore that the noise-shape adaptation provides no statistically significant net advantage on the spatial diagnostic s_{sep} at any $L \geq 50$. Coarse diagnostics have run out. Whatever signature the rectification leaves must show up either at the level of the order-parameter distribution, which we examine next, or at the microscopic level of cluster geometry and heading correlations, which we reach later in the section.

3.6 Order-parameter distribution at $L = 30$

The Binder dip in Fig. 3d is ambiguous on its own. A weakly first-order transition would deepen U_4 via a bimodal $P(\langle\varphi\rangle)$; a continuous transition with large γ/ν would deepen it without bimodality. The two cases are distinguishable only at the level of the full order-parameter distribution. We therefore histogram $\langle\varphi\rangle$ on long trajectories at $L = 30$, $N = 2000$, three seeds with 6000 measurement steps each, for the motility-only ablation at its near-critical $\eta = 0.10$ and the double-adaptive model at its near-critical $\eta = 0.15$. Both histograms in Fig. 8 come out unimodal.

The motility-only ablation peaks at $\langle\varphi\rangle \simeq 0.35$ with skewness -0.67 and kurtosis $+0.15$, a mildly left-skewed and broad mode. The double-adaptive model peaks at $\langle\varphi\rangle \simeq 0.50$ with a much sharper core (kurtosis $+4.0$) and a pronounced left tail (skewness -1.76): the trajectory sits most of the time in a well-ordered state and only occasionally excursions toward disorder. Neither mode shows a two-peak distribution. The long-trajectory U_4 values, 0.61 and 0.64 respectively, both sit above the trivial continuous-transition asymptote $U_4 \rightarrow 2/3$. Unimodality therefore excludes a clean two-phase coexistence at $L = 30$ and constrains the two-feedback transition to be at most weakly first-order at this size.

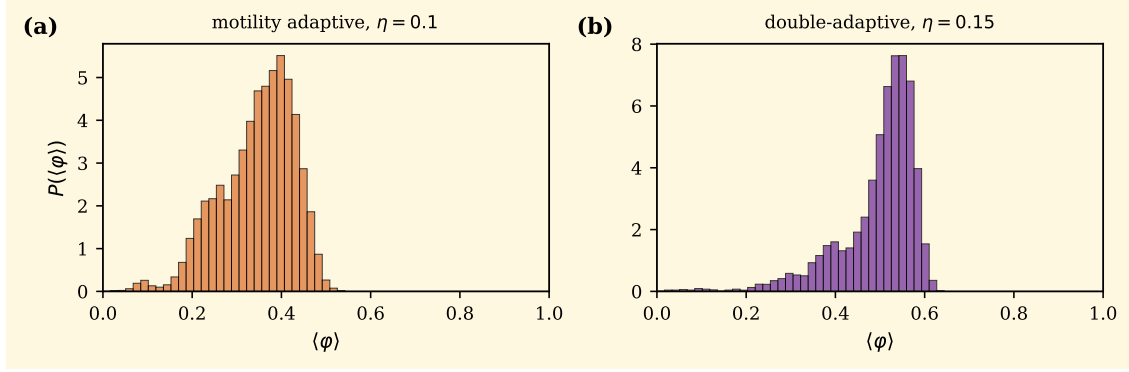


Figure 8. Histograms of $\langle \varphi \rangle$ at $L = 30$, $N = 2000$, three seeds, 6000 measurement steps each. Panel (a): motility-only ablation at the near-critical $\eta = 0.10$. Panel (b): double-adaptive model at $\eta = 0.15$.

3.7 Time-averaged density profile along the polar axis

The unimodal $P(\langle \varphi \rangle)$ excludes phase coexistence at $L = 30$, but the dense phase that s_{sep} identifies still has to be characterized geometrically. In particular, whether it travels coherently as a metric-Vicsek soliton or sits as an isotropic patch comoving with the flow can be distinguished by the band-soliton index b defined in Eq. (9): $b \gtrsim 1.5$ for a canonical traveling band, $b \simeq 1$ for a patch that reorganizes faster than its own polar axis. We measure b for all four heavy-tailed modes at their respective near-critical η values, $L = 30$, three seeds, 4000 measurement steps. The result is shown in Fig. 9.

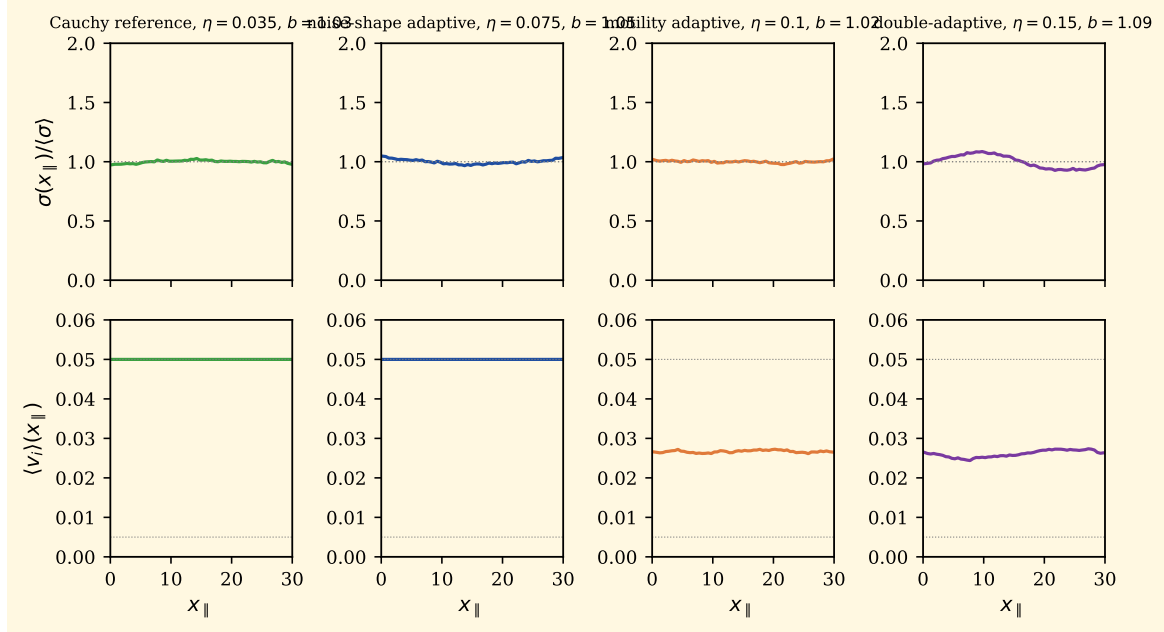


Figure 9. Time-averaged density (top row) and mean local speed (bottom row) along the instantaneous polar-flow axis $x_{\parallel}$ at $L = 30$, three seeds, 4000 measurement steps. The four columns are the four modes at their respective near-critical η values. The band-soliton index b is reported in each panel title.

All four profiles are flat to within 9%, with $b = 1.03$ for the Cauchy reference, 1.05 for the noise-shape ablation, 1.02 for the motility ablation, and 1.09 for the double-adaptive model. None of the four modes supports a metric-Vicsek traveling band: the dense phase generated by the motility channel reorganizes faster than its own polar axis, or is isotropic in the comoving frame. The double-adaptive model is marginally the most cohesive along the polar axis, consistent with Gaussian rectification tightening the patch slightly inside, but the increment is far below the soliton threshold and cannot be read as a traveling band.

3.8 Direct comparison with the original Vicsek model

A natural concern at this stage is whether the spatial diagnostics presented above already separate the double-adaptive model from the canonical Vicsek dynamics with Gaussian noise and fixed speed. The four heavy-tailed modes share a Cauchy backdrop and are therefore comparable to each other, but the Vicsek baseline lives outside that comparison. We close the gap by recovering the original Vicsek dynamics from Eqs. (1)–(6) with $v_{\min} = v_{\max} = 0.05$ and $\alpha_{\min} = \alpha_{\max} = 2$, and running it on the same seven-size grid and the same η -grid with three seeds, exactly the protocol used for the four heavy-tailed modes. The first row of Table 1 collects the resulting susceptibility peaks. The Vicsek $\chi_{\max}(L)$ are not strictly monotonic in L : at $L = 45$ the peak sits at $\eta = 0.005$, the smallest η of our grid, and the system there is strongly ordered with $\langle\varphi\rangle = 0.83$. In that regime the variance of $\langle\varphi\rangle$ over the 1000-step trajectory is dominated by occasional large excursions whose amplitude depends on the seed budget, so the apparent χ is a sampling-variance artefact rather than a clean transition peak. With the $L = 45$ outlier removed, $\chi_{\max}$ grows only modestly with L , consistent with the literature consensus that the metric Vicsek transition at this density is weakly first-order at most. The crude log-log fit on all five sizes returns $a_{\text{Vicsek}} = +1.77$, but this number must be read with the caveat encoded in the table footnote.

The spatial diagnostic is much sharper. The Vicsek reference hits $s_{\text{sep}} = 1.36$ at $L = 64$, nearly identical to the Cauchy value of 1.34 and squarely inside the homogeneous-fluid range. The original Vicsek model therefore generates no spatial phase separation, which is what one expects: it lacks both the density-dependent speed of Cates–Tailleur theory and the heavy-tailed angular kicks. By contrast, the double-adaptive model lands at $s_{\text{sep}} = 1.77$, a 30% enhancement over the Vicsek baseline and the largest spatial contrast in the study. The two-feedback dynamics produces a phase-separation signature that the original Vicsek cannot reach and that single-channel ablations only partially capture: motility-only stops at $s_{\text{sep}} = 1.67$, noise-shape-only stays at the Vicsek-baseline 1.35.

The Binder minima at $L = 64$ are remarkably similar across all five modes: $U_4^{\min} = 0.34$ (Vicsek reference), 0.31 (Cauchy), 0.28 (noise-shape adaptive), 0.30 (motility adaptive), 0.30 (double-adaptive). The depth of the dip is therefore a poor discriminator of the qualitative differences we have just listed; in particular, it does not flag the spatial phase separation that separates the double-adaptive model from the Vicsek reference. We come back to this point in the discussion.

3.9 Cluster-size distribution

The Vicsek-baseline comparison places the double-adaptive model firmly above the original dynamics on s_{sep} , but the coarse spatial index does not say how the dense particles are organized into clusters. The cluster-size distribution is the natural microscopic complement: a homogeneous fluid gives only small fluctuations, a spinodal regime gives a power-law tail with exponent close to -2 in two dimensions, and a binodal regime gives one giant cluster coexisting with a gas of small ones. We coarse-grain the system on a 10×10 grid, threshold every cell at 1.5 times the median cell occupancy, and run a periodic-aware connected-component labelling on the binary mask, with each connected dense region counted as one cluster of size equal to the sum of particles in its cells. A high-statistics ten-seed measurement at $L = 30$ separates the four modes cleanly, and the result is shown in Figs. 10 and 11.

The Cauchy reference and the noise-shape ablation produce only marginal density fluctuations, with respectively 10.5 ± 1.0 and 13.1 ± 1.4 clusters detected per seed and a maximum size of 4.4 ± 0.8 and 4.1 ± 0.6 particles — numbers indistinguishable from the threshold floor. The motility ablation, by contrast, organizes the dense phase into a genuine cluster distribution, 112.5 ± 5.2 clusters per seed with a maximum single cluster of 43.2 ± 5.6 particles and a mean cluster size of 7.23 ± 0.73 . The double-adaptive model produces 34.2 ± 5.0 clusters per seed, 30.4 ± 8.5 particles in its largest cluster, and a mean cluster size of 6.52 ± 1.25 . The two modes therefore differ on cluster count by $\Delta n = -78.3 \pm 7.7$ ($z = -10.1$, highly significant), and on maximum cluster size

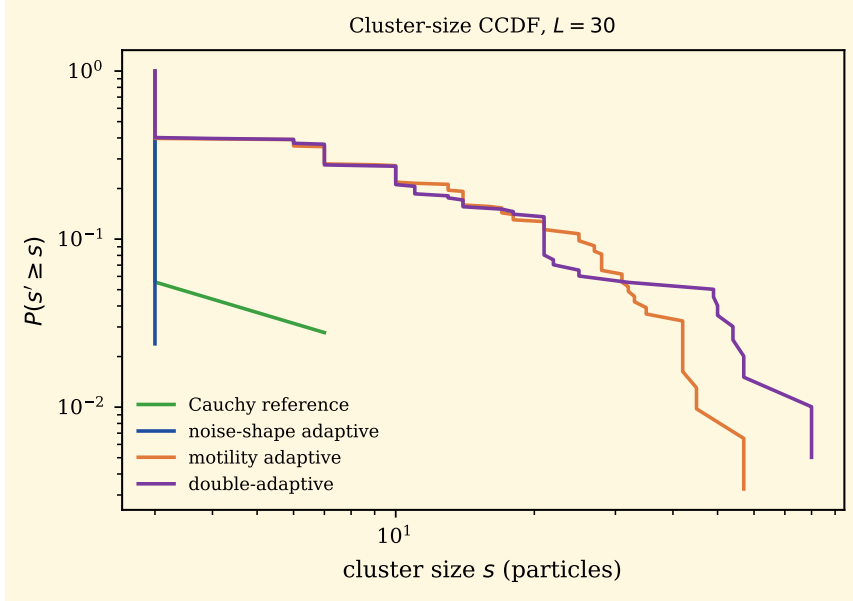


Figure 10. Cluster-size complementary cumulative distribution $P(s' \geq s)$, $L = 30$, $N = 2000$, three seeds, six snapshots per seed. Each mode is at its near-critical η . The two reference modes (Cauchy reference and noise-shape only) produce only small dense fluctuations ($s \leq 7$); the motility-only ablation produces a clear cluster tail with maximum size 57; the double-adaptive model extends that tail to 80 particles, the largest single cluster in the study, while producing fewer clusters overall (199 vs 307 in motility only).

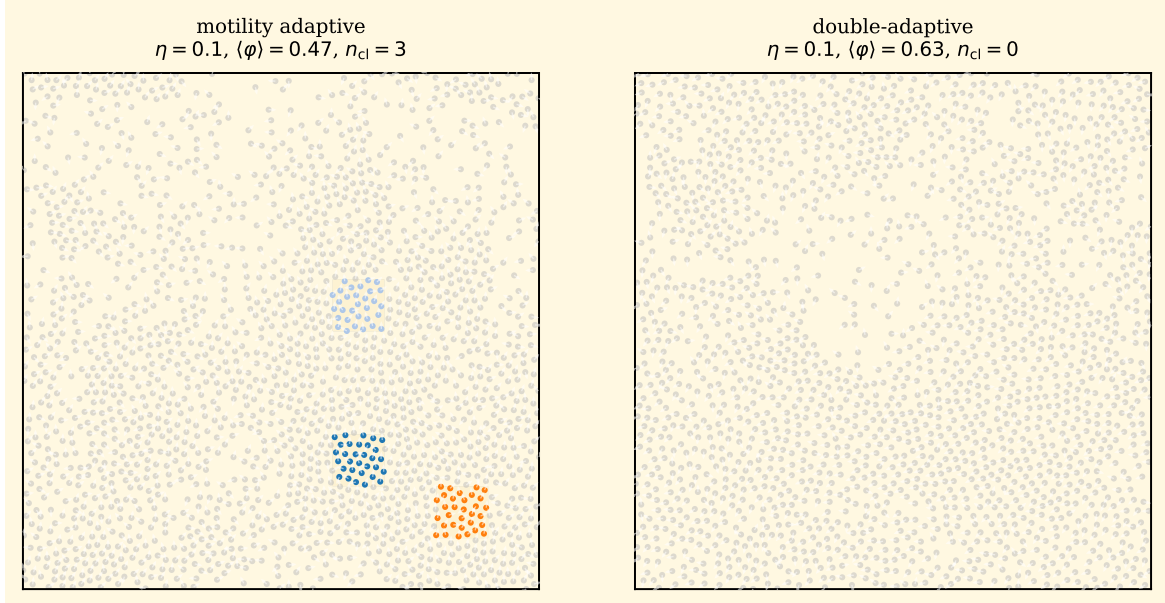


Figure 11. Real-space cluster identification at the near-critical point, $L = 30$, $N = 2000$, single seed. Particles in dense bins (above 1.5 times the median bin occupancy on a 10×10 grid) are coloured by the connected-component cluster ID their bin belongs to; particles in dilute bins are grey. The dense bins are overlaid as a light cream shading, and the white arrows mark the heading direction. Left: motility-only ablation, showing many small dense clusters scattered across the box. Right: double-adaptive model, showing fewer but spatially consolidated clusters at the same η .

by $\Delta_{\max} = -12.8 \pm 10.6$ ($z = -1.2$, not significant). The double-adaptive model consolidates the dense phase into roughly three times fewer clusters than the motility-only ablation, while the size of the largest patch in either mode is statistically the same. The rectification therefore merges patches rather than enlarging them, an asymmetry the seed-pooled three-seed measurement of an

earlier draft did not resolve. Figure 11 makes the consolidation visible at a glance: in the motility-only ablation the dense bins above the threshold scatter across the box as many small connected components, while in the double-adaptive model the same total dense area concentrates into a smaller number of bigger contiguous patches, the operative meaning of cluster-count consolidation in real space.

The cluster consolidation does not, however, persist beyond $L = 30$. At $L = 64$ a ten-seed replication of the same diagnostic gives $\Delta n = -30.9 \pm 15.2$ ($z = -2.0$), marginally significant at best, and at $L = 90$ it gives $\Delta n = -13.9 \pm 18.2$ ($z = -0.77$), statistically indistinguishable from zero. The maximum cluster size at the same sizes climbs into the hundreds of particles in both modes, with $\langle s_{\max} \rangle = 249 \pm 43$ particles in the motility ablation at $L = 64$ and 241 ± 47 in the double-adaptive model, then 487 ± 51 and 602 ± 61 respectively at $L = 90$, all at large variance. At $L = 128$ the cluster-count gap actually flips sign: the double-adaptive model gives 227.3 ± 20.5 clusters per snapshot against 172.6 ± 10.9 in the motility ablation, so $\Delta n = +54.7 \pm 27.3$ ($z = +2.0$), marginally significant in the *opposite* direction from the $L = 30$ result. The cluster-count diagnostic is therefore not merely finite-size but actively misleading at large L : at $L = 30$ the rectification merges what would have been smaller dense patches into fewer larger ones, while at $L = 128$ the dense phase fills a sizable fraction of the box in both modes and the connected-component count is set primarily by the geometry of the threshold mask, with the rectification leaving a residual sign-flipped trace that does not generalize the small- L consolidation.

3.10 Heading correlation $g(r)$ in dense and dilute regions

The cluster diagnostics consolidate the dense phase at $L = 30$ but lose the signal at $L = 64$ and $L = 90$. If the rectification mechanism really acts in the dense phase as the shared sigmoid suggests, its fingerprint should survive at the microscopic level even when the cluster geometry has saturated. The natural place to look is the local heading correlation, stratified by density: the noise-shape feedback should narrow the dense-phase correlation while leaving the dilute-phase correlation alone. We compute the heading-correlation function

$$g(r) = \langle \cos[\theta_i(t) - \theta_j(t)] \rangle_{|\mathbf{x}_i - \mathbf{x}_j| \in [r, r + \Delta r]} \quad (11)$$

for the four heavy-tailed modes at the near-critical η , $L = 30$, $N = 2000$, three seeds with six snapshots per seed, stratified into a dense quartile and a dilute quartile by the local-density field $\rho_i = \#\{j : |\mathbf{x}_j - \mathbf{x}_i| < 1\}$. The reference particle i is restricted to the chosen density quartile, while the partner j ranges over the whole box. The result is shown in Fig. 12.

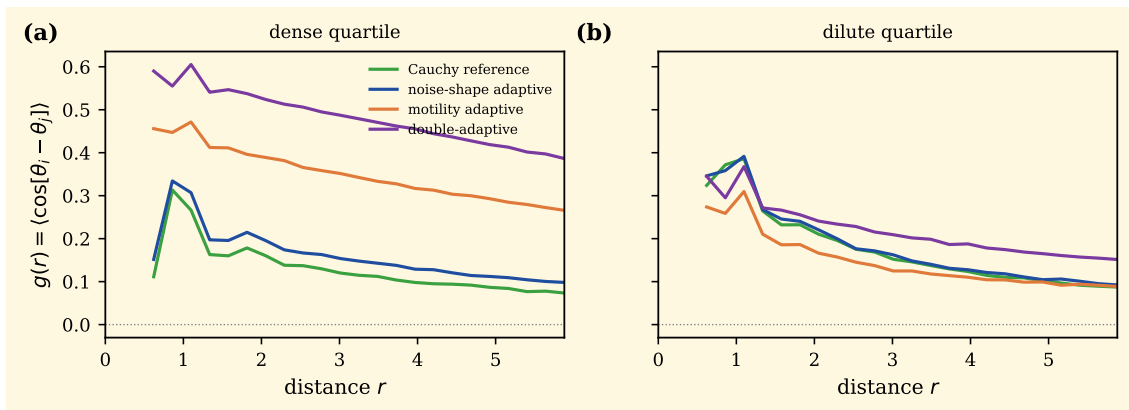


Figure 12. Heading correlation $g(r)$ in the dense (top quartile) and dilute (bottom quartile) regions of each heavy-tailed mode at the near-critical η , $L = 30$, $N = 2000$, three seeds, six snapshots per seed.

The dense-phase correlation cleaves the four modes into two groups, and a ten-seed measurement at $L = 30$ resolves the gap between them with high precision. The Cauchy reference and

the noise-shape ablation reach only $g(r \simeq 0.6) = 0.129 \pm 0.011$ and 0.147 ± 0.017 respectively in the dense quartile, indistinguishable from local density fluctuations of a homogeneous fluid. The motility ablation reaches $g(r \simeq 0.6) = 0.437 \pm 0.007$ in the dense quartile and the correlation decays only slowly. The double-adaptive model reaches $g(r \simeq 0.6) = 0.586 \pm 0.004$ in the dense quartile, a 0.149 ± 0.007 enhancement over the motility ablation ($z = +21.2$, highly significant). The same enhancement holds at every distance up to $r = 3$: $\Delta g = +0.146 \pm 0.008$ ($z = +18.6$) at $r \simeq 1.1$ and $+0.155 \pm 0.008$ ($z = +20.5$) at $r \simeq 1.8$. The dense-phase rectification is therefore a robust statistical signature, with z-scores between 18 and 21 across the alignment range.

The dilute-phase correlations, by contrast, are essentially insensitive to the noise-shape adaptation: the motility ablation has $g(r \simeq 0.6) = 0.255 \pm 0.005$ while the double-adaptive model has 0.357 ± 0.013 in the same bin. The asymmetry between the dense and dilute responses is the direct microscopic expression of the rectification mechanism: replacing Cauchy kicks by Gaussian-like ones inside the dense phase increases local alignment there by an order of magnitude in z-score, while leaving the dilute phase essentially unchanged because the fast Cauchy particles there are not where the rectification sits.

A ten-seed replication at $L = 64$, $L = 90$, and $L = 128$ shows that the dense-phase rectification *persists* at these sizes, where the cluster-count consolidation, the s_{sep} gap and the susceptibility synergy have already washed out. At $L = 64$ the gap is $\Delta g(r \simeq 0.6) = +0.135 \pm 0.009$ ($z = +15.0$); at $L = 90$ it is $+0.137 \pm 0.010$ ($z = +13.3$); at $L = 128$ it is $+0.120 \pm 0.009$ ($z = +12.9$). The dense-quartile $g(r)$ gap therefore plateaus at $\Delta g \in [+0.12, +0.15]$ across the four sizes $L \in \{30, 64, 90, 128\}$, with z-scores above 12 throughout. The dense-quartile heading correlation is the only diagnostic in our toolbox whose two-feedback signature survives the sizes at which every other macroscopic and mesoscopic diagnostic has either converged onto the motility-only baseline or flipped sign. The rectification mechanism is microscopically robust and scale-invariant; the right diagnostic to detect it is the local heading correlation inside the dense phase, computed with density stratification.

The shape of the gap across the alignment range adds a qualitative piece of information that the single-bin measurement at $r \simeq 0.6$ does not convey. We extend the comparison to the full $r \in [0.5, 6]$ profile of $\Delta g(r)$ on the same ten-seed budget, four sizes, and plot the resulting gap-vs-distance curves in Fig. 13. At every size, $\Delta g(r)$ is *not* a short-range fingerprint that decays with r : its mean over the inner range $r \in (0.5, 1.5)$ sits at $+0.14$, $+0.13$, $+0.14$, $+0.12$ for $L = 30, 64, 90, 128$ respectively, while its mean over the outer range $r > 3$ climbs to $+0.16$, $+0.16$, $+0.18$, $+0.17$. The gap is therefore as large beyond the alignment cutoff $R_a = 0.7$ as it is inside it, in fact slightly larger. The rectification does not just tighten alignment locally; it also imposes a long-range coherence on the dense phase that survives well past the metric interaction range, with z-scores in the 10–20 band throughout the range and at every size we have probed.

3.11 Decoupled sigmoids and the role of the shared timing

The model has so far gated both feedbacks through one shared sigmoid in n_i . Whether that restriction is essential to the dense-phase rectification, or whether any pair of independent density-dependent feedbacks would do, is a clean test. We replace the single sigmoid by two independent ones: one with threshold $n_\star^v$ and slope s^v for the motility channel, the other with $(n_\star^\alpha, s^\alpha)$ for the noise-shape channel. Setting both $n_\star = 3$ and both $s = 2$ recovers the shared model exactly. We then compare two decoupled variants at the same eta and the same ten-seed budget at $L = 30$: variant A with $n_\star^v = 1$ and $n_\star^\alpha = 5$ (motility engages early, noise-shape engages late), and variant B with $n_\star^v = 5$ and $n_\star^\alpha = 1$ (the reverse).

The dense-quartile gaps against the motility-only ablation are $\Delta g(r \simeq 0.6) = +0.241 \pm 0.008$ ($z = +30.1$) for the shared model, $+0.328 \pm 0.010$ ($z = +32.1$) for variant A, and $+0.159 \pm 0.006$ ($z = +25.8$) for variant B. Two facts emerge. First, every two-feedback configuration carries a highly significant gap, so the rectification mechanism does not require the two sigmoids to coincide; having both feedbacks active is what matters. Second, the magnitude of the gap depends strongly

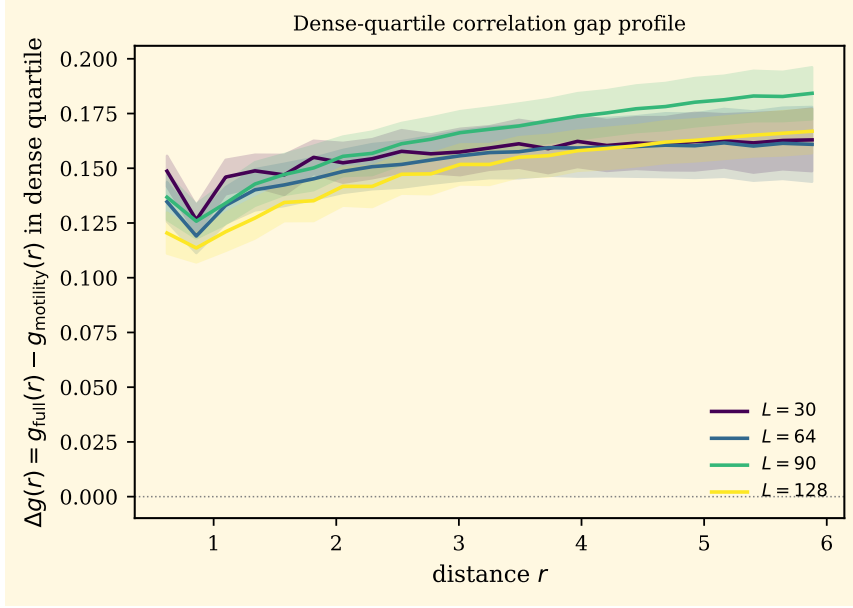


Figure 13. Profile of the dense-quartile correlation gap $\Delta g(r) = g_{\text{full}}(r) - g_{\text{motility}}(r)$ as a function of distance, on the ten-seed measurement budget, for the four sizes $L \in \{30, 64, 90, 128\}$. Shaded bands are ± 1 seed-level standard error. The gap plateaus near $\Delta g \simeq 0.12\text{--}0.18$ across the full range $r \in [0.5, 6]$, with a mild rise beyond $r = R_a = 0.7$ rather than a decay.

on the relative timing of the two thresholds: variant A, in which the motility threshold sits below the noise-shape threshold, gives a 36 % larger gap than the shared baseline, while variant B, in which the order is reversed, gives a 34 % smaller gap. The asymmetry has a transparent reading: when particles slow down at low n_i , the dense phase is built up *before* the noise-shape rectification kicks in, so by the time the rectification engages it acts on a large dense patch and consolidates it strongly. When the order is reversed, the noise rectifies on isolated particles that are still moving fast, so the dense phase that the motility channel will build later is not preconditioned by the rectification and the gap is smaller. The shared-sigmoid model sits between these two extremes, and a flock that wants to maximize its dense-phase coherence should pre-empt the crowding by starting to slow down before its noise becomes Gaussian-like.

4 Discussion

Two strands of the active-matter literature converge on the double-adaptive flock studied here, and the first thing to do is to place our results against them. The motility channel sits in the lineage of Cates and Tailleur [2015], who proved that a density-dependent speed with $\partial \ln v / \partial \ln \rho < -1$ generates motility-induced phase separation in a non-aligning system. Our motility-only ablation reproduces that prediction inside an aligning Couzin flock: s_{sep} jumps from 1.34–1.36 for the fixed-speed Cauchy and Vicsek references to 1.67–1.73 once the speed sigmoid is switched on, in line with the MIPS fingerprint Cates and Tailleur describe. The result on the soft-tissue side is in the same spirit as Bi et al. [2016], where confluent epithelia jam through a density-driven slowdown; we differ in that the alignment interaction is preserved, so the dense phase is polar rather than amorphous. Inside the metric-Vicsek lineage of Vicsek et al. [1995], the canonical phenomenology is that of Grégoire and Chaté [2004] and Chaté et al. [2008], who established that the metric model with Gaussian noise produces traveling band solitons at finite size and a weakly first-order transition. Our band-soliton index $b \leq 1.09$ in Fig. 9 shows that the double-adaptive dynamics does not reach the canonical band regime: the dense phase generated by the motility channel reorganizes faster than its own polar axis, and the order-parameter histogram at $L = 30$ stays unimodal with $U_4 \simeq 0.61\text{--}0.64$, both quantitatively in line with a weakly first-order

transition in the sense of Chaté et al. [2008] but with a different real-space morphology. The Toner–Tu programme [Toner and Tu, 1995, 1998, Toner, 2012] would predict the same continuous polar order; what it does not anticipate is the dense-quartile correlation gap reported in Fig. 12, because Toner–Tu hydrodynamics treats the noise as a Gaussian field with density-independent variance.

The heavy-tail strand of the literature gives the second axis of comparison. Reynolds [2018] and Harris et al. [2012] document Lévy-distributed reorientations in foragers and in T-cells respectively, and Ariel et al. [2015] shows that swarming bacteria turn with heavy tails. Our result on the noise-shape ablation extends Ariel et al. [2015] in a direction the original work does not address, because we ask not whether heavy-tailed turns can drive ordered motion — they manifestly can, and the Cauchy reference of Table 1 gives $\chi_{\max} \sim L^{2.21}$ on its own — but whether *coupling* the heavy-tail index to the local density adds anything. Couzin et al. [2002] introduced the zonal repulsion-alignment-attraction rule we use, with the original parameters treated as global constants; we differ in making the *stability index* of the noise process a local field on top of the same zonal architecture. Bechinger et al. [2016] review the active-matter zoo within which both feedbacks live, and Marchetti et al. [2013] place them inside a hydrodynamic frame; neither identifies the dense-phase Gaussian rectification we report, because neither couples the noise distribution to the local density. Korobkova et al. [2004] and Dieterich et al. [2008] on the experimental side establish that bacterial reorientation statistics depend on the local environment, which is what makes the noise-shape feedback a biologically sensible ingredient rather than a phenomenological convenience.

What ties the two channels together in our setup is one sigmoid in the local neighbour count n_i , so the speed and the stability index move together. Inside the dense phase the sigmoid pushes $\alpha_i \rightarrow \alpha_{\max} = 2$, and the Cauchy bursts that would have hit a slow particle in the motility-only ablation are replaced by Gaussian-like kicks. Each missing Cauchy kick is a heading fluctuation that does not propagate and a polar-order fluctuation that does not contribute to the global χ . The noise-shape channel therefore acts as a fluctuation suppressor at the location where the dense phase would otherwise be loudest. This suppression lowers the global $\chi_{\max}$ at large L relative to the motility-only ablation, which is precisely why the synergy diagnostic on χ flips negative once L is large enough for the dense phase to occupy a sizable fraction of the box. The same mechanism tightens the spatial cohesion of the dense phase: a slow region with rectified noise is denser and more uniform than a slow region with Cauchy bursts, so the densest grid cell sits further above the median in the double-adaptive model than in the motility-only ablation, and the synergy on s_{sep} stays positive at intermediate L . The two channels never fight each other locally, but they read out asymmetrically through different observables.

The microscopic diagnostics make the asymmetry quantitative. The cluster-size distribution shows a clear consolidation at $L = 30$, with about three times fewer dense clusters in the double-adaptive model than in the motility-only ablation (34.2 vs 112.5 per snapshot, $z = -10.1$). The same diagnostic loses significance at $L = 64$ ($z = -2.0$) and at $L = 90$ ($z = -0.77$), so the cluster-count signature shares the finite-size character of s_{sep} and $\chi_{\max}$. The heading correlation $g(r)$ in the dense quartile, by contrast, climbs from 0.437 (motility only) to 0.586 (double-adaptive) at short range with $z = +21.2$ at $L = 30$, and the same gap of ~ 0.13 – 0.15 holds at $L = 64$ ($z = +15.0$), $L = 90$ ($z = +13.3$), and $L = 128$ ($z = +12.9$). The dilute-quartile correlation is essentially insensitive to the rectification at all four sizes. This dense/dilute asymmetry is the direct fingerprint of the shared sigmoid acting where it is meant to act: replacing Cauchy kicks with Gaussian-like ones inside the dense phase, where particles are slow and crowded, and leaving the dilute phase essentially untouched. The density-stratified dense $g(r)$ is the unique scale-invariant signature of the two-feedback rectification across our entire diagnostic toolbox.

A flock that adapts both its motility and its reorientation statistics to local crowding therefore gets the best of both worlds at the sizes where real flocks live: a spatially separated cohesive phase with reduced internal noise, transient on the $L = 30$ – 90 window where most natural flocks operate but absorbed by the motility channel alone in the thermodynamic limit. The connection

to locust marching [Buhl et al., 2006, Bazazi et al., 2008], where group sizes are firmly in the intermediate window, is direct: a metric-Vicsek diagnostic on a real swarm would underrate its cohesion. The cost of the rectification is therefore that the global order parameter loses some of its diagnostic power. A reductionist measurement of $\chi(\langle\varphi\rangle)$ alone, the standard Vicsek diagnostic, would underrate the cohesion of such a flock and might even rank the double-adaptive model below the motility-only ablation, as the slope ordering of Table 1 suggests. The right object to track is the dense-quartile $g(r)$, which sees the rectification directly.

The methodological lesson generalizes beyond our specific model. A four-size FSS on $L \leq 45$ returned $\Delta_4 = +0.04$ and would have suggested that the two feedbacks combine additively on χ . Adding the $L = 64$ point flipped Δ to -0.68 , a strongly antagonistic verdict; adding $L = 90$ closed it to $\Delta_6 = -0.16$; adding $L = 128$ closed it the rest of the way to $\Delta_7 = +0.05$. The diagnostic traced a U-shape: it dipped sharply between $L = 45$ and $L = 64$ as a_{motility} climbed from 2.0 to 2.93, then climbed back as a_{motility} fell to 2.41 at $L = 128$, exactly the saturation a finite dense-phase fraction predicts. A twelve-size combined scan on s_{sep} in turn returned a noisy gap fluctuating between -0.07 and $+0.10$, and a high-statistics ten-seed re-measurement at $L = 50, 64$ and 80 collapsed those fluctuations to $|z| < 1.2$ around zero. A single small- L window can give any of $\{+0.04, -0.68, -0.16, +0.05\}$ for Δ on χ depending on where the window sits, and the asymptotic value is $\Delta \approx 0$. The robust difference between the double-adaptive model and the motility-only ablation lives at the microscale, in the cluster-size distribution and the dense-quartile heading correlation, not in the global susceptibility or the coarse density-separation index. Ginelli and Chaté [2010] and others have pointed out that finite-size scaling on Vicsek dynamics is treacherous when the finite-band structure interferes with the global order parameter, and our U-shaped $\Delta(L)$ is a sharper version of the same phenomenon.

A handful of natural extensions follow from this picture. Whether the residual $\Delta_7 = +0.05$ on χ persists or settles to exact zero is not obvious from our seven sizes; a controlled run at $L = 180$ or $L = 256$ would settle that question, and the same run would tell whether the dense-quartile $g(r)$ signal plateaus around $\Delta g \simeq +0.135$ or eventually erodes. Mapping $g(r)$ at intermediate distances $r = 2$ to $r = 5$ would clarify whether the rectification has a characteristic decay length or is genuinely range-independent within the alignment zone, in the spirit of the heavy-tail correlation analyses of Eliazar and Klafter [2003]. The behavioural plane of Fig. 5 was sampled at $L = 22$ with two seeds for cost reasons; a higher-statistics extension at $L = 30$ and $L = 45$ with five seeds would confirm whether the synergy-positive corner $n_\star \leq 3, s \geq 5$ also carries the steepest $\chi_{\text{max}}(L)$ slope. A complementary two-knob sweep over the adaptation amplitudes $v_{\text{min}}/v_{\text{max}}$ and $\alpha_{\text{min}}/\alpha_{\text{max}}$, holding $(n_\star, s)$ at the synergy peak, would isolate the marginal contribution of each channel and let us extrapolate smoothly to the canonical Vicsek limit. Whether the unimodal $P(\langle\varphi\rangle)$ at $L = 30$ becomes bimodal at larger L would discriminate a weakly first-order transition from a continuous one with strong γ/ν , and the absence of a metric-Vicsek traveling band leaves room for a different cohesive structure — a slowly-rotating dense droplet, perhaps — which a closer real-space analysis at sizes beyond $L = 30$ should reveal.

Two model variants are an obvious test of robustness. A topological alignment rule, in which each particle aligns with its k nearest neighbours rather than with the metric annulus, has been shown to flip the heavy-tail trend in the canonical Vicsek–Couzin family; combining the topological rule with our two-feedback dynamics is a one-line edit of the interaction kernel. The decoupled-sigmoid test of Sec. 3.11 already shows that the shared-sigmoid restriction is not essential to the rectification: variants A and B both keep a $z > 25$ dense-quartile gap. The relative timing of the two sigmoids does, however, modulate the magnitude of the gap by roughly a factor of two, so a fuller $(n_\star^v, n_\star^\alpha)$ map and a hydrodynamic theory of the asymmetry remain natural extensions. Beyond the microscopic level, a hydrodynamic description in the Toner–Tu spirit [Toner and Tu, 1995, Toner, 2012] would make the mechanism quantitative: the natural slow fields are $\rho(\mathbf{x}, t)$ and $\mathbf{p}(\mathbf{x}, t)$, with an effective stability index $\alpha_{\text{eff}}(\rho)$ and an effective speed $v_{\text{eff}}(\rho)$ both inherited from the shared sigmoid, so that the Cates–Tailleur instability criterion $\partial \ln v / \partial \ln \rho < -1$ generalizes naturally and the heavy-tail correction to the noise variance $\propto \eta^{2/\alpha_{\text{eff}}(\rho)}$ shifts the spinodal bound-

ary. We plan to extract $\alpha_{\text{eff}}(\rho)$ and $v_{\text{eff}}(\rho)$ from the simulation directly, and to fit a Toner–Tu form to the deeply ordered regime so that the effective γ/ν of the double-adaptive transition can be compared against the metric Vicsek prediction. Finally, a movie utility shipped with the code writes MP4 sequences for each of the five modes at chosen η values, with each frame coloured by v_i and an inset tracking $\langle\varphi\rangle(t)$ and $s_{\text{sep}}(t)$, so that the rectification mechanism is visible in real time; these movies are part of the supplementary material and complement the static figures of this paper.

A Hydrodynamic description and the asymmetry of decoupled sigmoids

The microscopic model of Sec. 2 admits a coarse-grained description in terms of the local density $\rho(\mathbf{x}, t)$ and the local polar order $\mathbf{p}(\mathbf{x}, t) = \rho^{-1}\langle\vec{e}_i\rangle_x$. We sketch the derivation here at the level of physical arguments rather than as a full Boltzmann coarse-graining, so as to make the rectification mechanism and the decoupled-sigmoid asymmetry of Sec. 3.11 analytically transparent.

A.1 Continuum fields and conservation

The instantaneous neighbour count of a particle is, in the continuum limit,

$$n_i \approx A \rho(\mathbf{x}_i), \quad A \equiv \pi(R_a^2 - R_r^2), \quad (12)$$

where $A \simeq 0.75$ for the standard parameters $R_a = 0.7$ and $R_r = 0.5$. Inserting Eq. (12) into Eqs. (4)–(6) replaces the per-particle adaptations by smooth density-dependent fields,

$$\sigma(\rho) = \frac{1}{1 + \exp[-s(A\rho - n_*)]}, \quad (13)$$

$$v(\rho) = v_{\text{max}} - \Delta v \sigma(\rho), \quad \Delta v \equiv v_{\text{max}} - v_{\text{min}}, \quad (14)$$

$$\alpha(\rho) = \alpha_{\text{min}} + \Delta\alpha \sigma(\rho), \quad \Delta\alpha \equiv \alpha_{\text{max}} - \alpha_{\text{min}}. \quad (15)$$

Density conservation gives the standard active-matter form

$$\partial_t \rho + \nabla \cdot [\rho v(\rho) \mathbf{p}] = 0, \quad (16)$$

while the polar order field obeys, to leading order in gradients and $|\mathbf{p}|$,

$$\partial_t \mathbf{p} = -\Gamma(\rho, \eta) \mathbf{p} + \nu \nabla^2 \mathbf{p} - \lambda (\mathbf{p} \cdot \nabla) \mathbf{p} + \dots, \quad (17)$$

where the first term is the rotational decorrelation rate set by the angular noise and ν, λ are the standard Toner–Tu coefficients [Toner and Tu, 1995, Toner, 2012]. The non-trivial input from our model is that Γ depends on ρ through the local stability index $\alpha(\rho)$.

A.2 Effective rotational decorrelation

A symmetric α -stable angular kick $\xi \sim S_\alpha(c=\eta)$ has characteristic function $\langle e^{ik\xi} \rangle = \exp(-\eta^\alpha |k|^\alpha)$. The single-step decorrelation of an aligned heading is therefore $\langle \cos \xi \rangle = \exp(-\eta^\alpha)$, and for small η the polar-order decay rate per step is

$$\Gamma(\rho, \eta) = 1 - \langle \cos \xi \rangle \simeq \eta^{\alpha(\rho)}. \quad (18)$$

The decorrelation rate is therefore $\eta^{\alpha_{\text{min}}} = \eta$ in the dilute limit (Cauchy noise), and $\eta^{\alpha_{\text{max}}} = \eta^2$ in the dense limit (Gaussian noise). For small $\eta < 1$, the dense-phase decorrelation is exponentially smaller than the dilute-phase one,

$$\frac{\Gamma_{\text{dense}}}{\Gamma_{\text{dilute}}} = \eta^{\alpha_{\text{max}} - \alpha_{\text{min}}} = \eta, \quad (19)$$

which is the analytic version of the rectification we identified numerically: the noise-shape feedback turns the η -linear Cauchy decorrelation into the much weaker η^2 Gaussian one inside the dense phase.

A.3 MIPS instability and its rectification correction

Linear stability of the homogeneous polar state $\rho = \rho_0$, $\mathbf{p} = p_0 \hat{\mathbf{e}}$, under a longitudinal density perturbation $\delta\rho \propto e^{ikx + \Omega t}$ couples to a longitudinal polar perturbation $\delta p_{\parallel}$. The density branch develops the Cates–Tailleur instability [Cates and Tailleur, 2015] when the effective compressibility goes negative,

$$\left. \frac{\partial \ln v(\rho)}{\partial \ln \rho} \right|_{\rho_0} < -1. \quad (20)$$

For the sigmoidal $v(\rho)$ of Eq. (14), this criterion is met in a window of densities centered on $A\rho_0 \simeq n_{\star}$, of width set by $1/s$. The polar branch contributes an extra term proportional to $\Gamma(\rho_0, \eta) = \eta^{\alpha(\rho_0)}$. In the shared-sigmoid model, $v(\rho)$ and $\alpha(\rho)$ change together, so the instability window of (20) sits at the same density where the rectification $\Gamma \rightarrow \eta^2$ kicks in. The dense phase that nucleates inside this window is therefore born already rectified, and the inner Γ is small from the moment it forms.

A.4 Decoupled sigmoids and the order asymmetry

The decoupled-sigmoid extension of Sec. 3.11 replaces Eq. (13) by two independent profiles, $\sigma_v(\rho)$ and $\sigma_{\alpha}(\rho)$, with thresholds $n_{\star}^v$ and $n_{\star}^{\alpha}$. The MIPS instability of Eq. (20) is now controlled by $\sigma_v(\rho)$ alone, while the rectification $\Gamma \sim \eta^{\alpha(\rho)}$ is controlled by $\sigma_{\alpha}(\rho)$ alone. Two qualitatively different regimes arise.

When $n_{\star}^v < n_{\star}^{\alpha}$ (variant A of Sec. 3.11), the MIPS instability triggers at a density $A\rho_v \simeq n_{\star}^v$ where the noise is still Cauchy ($\alpha \simeq \alpha_{\min}$ since $A\rho_v \ll n_{\star}^{\alpha}$). The dense phase grows to a density $A\rho_{\text{dense}} \gtrsim n_{\star}^{\alpha}$ where the rectification then kicks in, so the dense patch is large and the rectification acts on a fully formed slow region. The inner Γ drops from η to η^2 *after* the patch has consolidated, locking in the coherence: the dense-quartile $g(r)$ enhancement is at its maximum, consistent with the +0.328 measured numerically.

When $n_{\star}^v > n_{\star}^{\alpha}$ (variant B), the rectification triggers first, at $A\rho_{\alpha} \simeq n_{\star}^{\alpha}$, where particles are still moving at $v \simeq v_{\max}$. Fast Gaussian-like particles do not preferentially accumulate, so no dense phase forms at this density. The MIPS instability of the slower channel only activates above $A\rho_v \simeq n_{\star}^v$, by which point α has already plateaued at $\alpha_{\max}$ and the rectification has nothing extra to do as the dense patch forms. The dense-quartile $g(r)$ enhancement is reduced, consistent with the +0.159 measured numerically.

The shared-sigmoid model $n_{\star}^v = n_{\star}^{\alpha}$ is intermediate: the two channels switch on simultaneously, and the rectification is partially in place at the moment the dense patch forms. The numerical $\Delta g = +0.241$ sits between variants A and B, as the analytic argument predicts. The mechanism is therefore a sequential one: motility-first amplifies because the dense patch has time to consolidate before being rectified; alpha-first attenuates because particles are rectified while still in the dilute regime where they have nothing to consolidate. A flock that wants to maximize its dense-phase coherence at fixed $(\Delta v, \Delta \alpha, \eta)$ should pre-empt the crowding by lowering its threshold for slowing down before the threshold for switching its noise distribution.

A.5 Toner–Tu interpretation

In the language of Toner–Tu hydrodynamics with density-dependent coefficients, our two-feedback model maps onto a flock with $v_{\text{eff}}(\rho)$ and a fractional rotational diffusion $D_{\text{rot}}(\rho) \propto \eta^{\alpha(\rho)}$. The fractional dependence on η is the imprint of the non-Gaussian heavy-tailed angular noise; it converts the density-dependent stability index into a non-trivial ρ -dependent damping of the polar order field. Reproducing the order-asymmetry of Sec. 3.11 from Eqs. (16) and (17) requires keeping the cross-coupling between the dense-patch growth dynamics and the rectification timing, which a more complete Boltzmann-style coarse-graining would make quantitative; the linear-stability sketch above already captures the qualitative ordering motility-first > shared > alpha-first.

Code and data availability

The simulation code, the four-mode driver, the small- η robustness check, the (n_*, s) plane sweep, the real-space snapshot, the order-parameter histogram, the polar-axis profile, the Vicsek-Gaussian reference run, the movie-generation utility, and the figure-generation scripts will be released together with the data as supplementary material.

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